

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Practices Description

Degree, Semester& Branch: VI Semester B.E. ECE B

Course Code & Title: EC8095 & VLSI Design

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: MOS Transistor

Name of the Innovative Practice: One minute paper

Date & Time: 17.12.2019 & 5 minutes

ICT Tool Used:- Smart Class Room

Description of One Minute Paper Activity:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

Goals (Learning Outcomes):

1. The students will be able to acquire better knowledge about Metal Oxide Semiconductor (MOS) Transistor.
2. The students will be able to understand the different regions of the MOS Transistor

Use of appropriate methods:

Justification for choosing the One Minute Paper Activity:

Students already studied Metal Oxide Semiconductor Field Effect Transistor (MOSFET) in Electron Devices subject. However, I have reviewed about it. I want to know the knowledge level of the students about MOSFET. For that, I have chosen One Minute Paper as an Innovative practice method. In that I have asked some questions related with MOSFET.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution	
MOS Transistor	
	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was assessed by evaluating the one minute paper and its feedback was given to the students in the next class.
- ❖ The assessment of effectiveness of the activity was felt when most of the students eagerly participated and they have written MOS transistor operations.

Reflective Critique:

- **Benefits:**

The students confused about the operating regions of the MOSFET. After that, I have reviewed about MOS transistor. Then students realized the behaviour of MOS transistor and most of the students have answered properly in the one minute paper.

- **Challenges:**

Initially, I have planned the activity for 2 minutes but students are confused about the operating regions of the MOSFET and then I have reviewed it which takes another 3 minutes for completing the activity.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1					2	1		

CO1: The Students will be able to realize the concepts of digital building blocks using MOS transistor.

References:

1. Neil H.E. Weste, David Money Harris “CMOS VLSI Design: A Circuits and Systems Perspective”, 4th Edition, Pearson, 2017.
2. <https://provost.tufts.edu/celt/files/MinutePaper.pdf>

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Course Code & Title: EC8095 & VLSI Design

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Stick Diagrams

Name of the Innovative Practice: Video Lecture

Date & Time: 02.01.2020 & 10 minutes

ICT Tool Used:- LCD Projector

Description of Video Lecture:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

1. The students will be able to analyze need for stick diagrams.
2. The students will be able to understand static CMOS logic and Layout design rules.

Use of appropriate methods:

Justification for choosing the Video Lecture:

Some of the students confused about Stick diagrams and Layout diagrams. For that, I have chosen Stick diagram video lecture which enable them to differentiate stick diagram and Layout diagram. And also it gives the significance of the stick diagram.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution	
Stick Diagrams	
	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was assessed by asking questions to some of the students who are not shown interest towards the subject and they are answered.
- ❖ The assessment of effectiveness of the activity was felt when they have listened the class effectively.

Reflective Critique:

- **Benefits:**

Some of the students confused about Stick diagram, Layout design rules and Layout diagram. So that, I have chosen video lecture for better understanding about Stick diagram, Layout design rules and Layout diagram. After shown this video, Students know the significance of Stick diagram, Layout design rules and Layout diagram.

- **Challenges:**

Initially, I have planned to show the video lecture for 5 minutes but initially the audio quality was poor. So I have adjusted the amplifier for getting clear sound which it takes another 5 minutes for completing the activity.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1					2	1		

CO1: The Students will be able to realize the concepts of digital building blocks using MOS transistor.

References:

1. Neil H.E. Weste, David Money Harris “CMOS VLSI Design: A Circuits and Systems Perspective”, 4th Edition, Pearson, 2017.
2. <https://www.youtube.com/watch?v=wqRGa5sOUmc>